

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Matrix Switcher
Manufacturer: Extron
Firmware Version: 2.01
Model(s): DMS 3600, DMS 1600, DMS 3200, DMS 2000

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
2.08.0001-b001	1.4.2

Version History

Module Version	Date	Notes
1_3_0_1	3/28/2018	Fixed status for RefreshMatrix command
1_3_0_0	3/1/2018	Added EDIDAssignment and RefreshMatrix commands. Renamed InputSignal to InputSignalStatus and updated its state from 'Inactive' to 'Not Active'. Added Update method support for InputTieStatus and OutputTieStatus. Updated to latest Matrix standard.
1_1_1_0	10/3/2017	Updated comm sheet to Rev. B
1_1_0_0	2/24/2017	Fixed error checking. Removed Tie Type qualifier from OutputTieStatus command. Fixed InputSignal Status, InputTieStatus, and OutputTieStatus.

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1_0_3_3	1/31/2017	Updated verbose mode logic.
1_0_3_2	7/25/2016	Update to standard.
1_0_3_1	2/26/2016	Fixed update issue.
1_0_3_0	2/12/2016	Initial Version.

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`
- If login credentials are required, devicePassword must be set accordingly.
Example: `InterfaceName.devicePassword = 'extron'`

Supported Classes and Examples

SerialClass
<code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='DMS 3600')</code>
SerialOverEthernetClass
<code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='DMS 3600')</code>
EthernetClass
<code>InterfaceName = ModuleName.EthernetClass('192.168.254.254', 23, Model='DMS 3600')</code>

Set Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command ExecutiveMode	Value 'On'	Value 'Off'
# ExecutiveMode example InterfaceName.Set('ExecutiveMode', 'On')		
Command GlobalVideoMute	Value 'On'	Value 'Off'
# GlobalVideoMute example InterfaceName.Set('GlobalVideoMute', 'On')		
Command MatrixTieCommand	Value None	
Qualifier Key 'Input'	Qualifier Value '1' – '36' ²	
Qualifier Key 'Output'	Qualifier Value '1' – '36' ¹	Qualifier Value 'All'
# MatrixTieCommand example InterfaceName.Set('MatrixTieCommand', None, {'Input': '1', 'Output': '1'})		
Command PresetRecall	Value '1' – '32'	
# PresetRecall example InterfaceName.Set('PresetRecall', '1')		
Command PresetSave	Value '1' – '32'	
# PresetSave example InterfaceName.Set('PresetSave', '1')		
Command RefreshMatrix	Value '1 - 16' 'All'	Value '17 - 32' ⁵ '17 - 20' ³ Value '33 - 36' ⁴
# RefreshMatrix example InterfaceName.Set('RefreshMatrix', '1 - 16')		
Command VideoMute	Value 'On'	Value 'Off'
Qualifier Key 'Output'	Qualifier Value '1' – '36' ¹	
# VideoMute example		

```
InterfaceName.Set('VideoMute', 'On', {'Output': '1'})
```

¹ 'Output' qualifier range is proportional to the number of outputs on the device.

² 'Input' qualifier range is proportional to the number of inputs on the device.

³ Only available for DMS 2000

⁴ Only available for DMS 3600

⁵ Only available for DMS 3200 and DMS 3600

Status Available

For all commands except for InputSignalStatus, Update should be called only once since the command's status will be updated automatically as the device's status changes. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command	Value	Value	
ConnectionStatus	'Connected'	'Disconnected'	
# ConnectionStatus examples Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command	Value	Value	Value
EDIDAssignment	'Output 1'	'Output 2'	'Output 3'
	'Output 4'	'Output 5'	'Output 6'
	'Output 7'	'Output 8'	'Output 9'
	'Output 10'	'Output 11'	'Output 12'
	'Output 13'	'Output 14'	'Output 15'
	'Output 16'	'Output 17'	'Output 18'
	'Output 19'	'Output 20'	'Output 21'
	'Output 22'	'Output 23'	'Output 24'
	'Output 25'	'Output 26'	'Output 27'
	'Output 28'	'Output 29'	'Output 30'
	'Output 31'	'Output 32'	'Output 33'
	'Output 34'	'Output 35'	'Output 36'
	'640x480 @ 60Hz'	'640x480 @ 75Hz'	'800x600 @ 60Hz'
	'800x600 @ 75Hz'	'852x480 @ 60Hz'	'852x480 @ 75Hz'
	'1024x768 @ 60Hz'	'1024x768 @ 75Hz'	'1024x852 @ 60Hz'
	'1024x852 @ 75Hz'	'1280x768 @ 60Hz'	'1280x768 @ 75Hz'
	'1280x1024 @ 60Hz'	'1280x1024 @ 75Hz'	'1365x768 @ 60Hz'

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	'1365x768 @ 75Hz'	'1366x768 @ 60Hz'	'1366x768 @ 75Hz'
	'1400x1050 @ 60Hz'	'1600x1200 @ 60Hz'	'1920x1200 @ 60Hz'
	'480p @ 60Hz'	'576p @ 50Hz'	'720p @ 50Hz'
	'720p @ 60Hz'	'1080p @ 60Hz'	'1080i @ 60Hz'
	'1080p @ 50Hz'	'1080p @ 60Hz, stereo'	'User Assigned 1'
	'User Assigned 2'	'User Assigned 3'	'User Assigned 4'
	'User Assigned 5'	'User Assigned 6'	'User Assigned 7'
	'User Assigned 8'		
Qualifier Key 'Input'	Qualifier Value '1' – '36' ²		
# EDIDAssignment examples InterfaceName.Update('EDIDAssignment', {'Input': '1'}) Value = InterfaceName.ReadStatus('EDIDAssignment', {'Input': '1'}) InterfaceName.SubscribeStatus('EDIDAssignment', None, FeedbackHandler)			
Command ExecutiveMode	Value 'On'	Value 'Off'	
# ExecutiveMode examples InterfaceName.Update('ExecutiveMode') Value = InterfaceName.ReadStatus('ExecutiveMode') InterfaceName.SubscribeStatus('ExecutiveMode', None, FeedbackHandler)			
Command InputSignalStatus ⁴	Value 'Active'	Value 'Not Active'	
Qualifier Key 'Input'	Qualifier Value '1' – '36' ²		
# InputSignalStatus examples InterfaceName.Update('InputSignalStatus', {'Input': '1'}) Value = InterfaceName.ReadStatus('InputSignalStatus', {'Input': '1'}) InterfaceName.SubscribeStatus('InputSignalStatus', None, FeedbackHandler)			
Command InputTieStatus	Value 'Video'	Value 'Untied'	
Qualifier Key 'Input'	Qualifier Value '1' – '36' ²		
Qualifier Key 'Output'	Qualifier Value '1' – '36' ¹		
# InputTieStatus examples InterfaceName.Update('InputTieStatus', {'Input': '1', 'Output': '1'}) Value = InterfaceName.ReadStatus('InputTieStatus', {'Input': '1', 'Output': '1'}) InterfaceName.SubscribeStatus('InputTieStatus', None, FeedbackHandler)			
Command OutputTieStatus	Value '1' – '36' ³		
Qualifier Key 'Output'	Qualifier Value '1' – '36' ¹		

OutputTieStatus examples

InterfaceName.Update('OutputTieStatus', {'Output': '1'})

Value = InterfaceName.ReadStatus('OutputTieStatus', {'Output': '1'})

InterfaceName.SubscribeStatus('OutputTieStatus', None, FeedbackHandler)

Command	Value	Value
VideoMute	'On'	'Off'
Qualifier Key	Qualifier Value	
'Output'	'1' – '36' ¹	
# VideoMute examples		
InterfaceName.Update('VideoMute', {'Output': '1'})		
Value = InterfaceName.ReadStatus('VideoMute', {'Output': '1'})		
InterfaceName.SubscribeStatus('VideoMute', None, FeedbackHandler)		

¹ 'Output' qualifier range is proportional to the number of outputs on the device.² 'Input' qualifier range is proportional to the number of inputs on the device.³ OutputTieStatus range is proportional to the number of inputs on the device.⁴ Calling Update method with any one qualifier value will update statuses for all other qualifier values.

Cable and Adapter Requirements

Captive Screw to Male DB9 RS-232 Serial Cable

Notes for the Device

Serial communication

Port Type: RS-232

Baud Rate: 9600

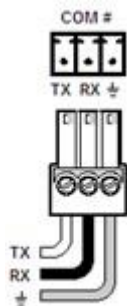
Data Bits: 8

Parity: None

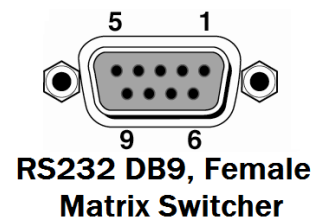
Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD		2	TxD
RxD		3	RxD
GND		5	GND



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scripter ethernet interface

Port Type:	Ethernet
Default Port:	23
Logon Credentials Supported:	Yes
Multi-Connection	Yes
Capabilities:	
Port Changeability:	Yes

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication.
Control of the device may be limited if "User" password is used for authentication.

Notes for the Device

Appendix A. Set Commands

Executive Mode Off	0X
Executive Mode On	1X
Global Video Mute Off	0*B
Global Video Mute On	1*B
Matrix Tie Command None Output 1 Input 0	0*1&
Matrix Tie Command None Output 32 Input 0	0*32&
Matrix Tie Command None Output All Input 0	0*1&0*2&0*3&0*4&0*5&0*6&0*7&0*8&0*9&0*10&0*11&0*12&0*13&0*14&0*15&0*16&0*17&0*18&0*19&0*20&0*21&0*22&0*23&0*24&0*25&0*26&0*27&0*28&0*29&0*30&0*31&0*32&0*33&0*34&0*35&0*36&
Matrix Tie Command None Output 1 Input 32	32*1&
Matrix Tie Command None Output 32 Input 32	32*32&
Matrix Tie Command None Output All Input 32	32*1&32*2&32*3&32*4&32*5&32*6&32*7&32*8&32*9&32*10&32*11&32*12&32*13&32*14&32*15&32*16&32*17&32*18&32*19&32*20&32*21&32*22&32*23&32*24&32*25&32*26&32*27&32*28&32*29&32*30&32*31&32*32&32*33&32*34&32*35&32*36&
Matrix Tie Command None Output 20 Input 0	0*20&
Matrix Tie Command None Output 1 Input 20	20*1&
Matrix Tie Command None Output 20 Input 20	20*20&
Matrix Tie Command None Output All Input 20	20*1&20*2&20*3&20*4&20*5&20*6&20*7&20*8&20*9&20*10&20*11&20*12&20*13&20*14&20*15&20*16&20*17&20*18&20*19&20*20&20*21&20*22&20*23&20*24&20*25&20*26&20*27&20*28&20*29&20*30&20*31&20*32&20*33&20*34&20*35&20*36&
Matrix Tie Command None Output 16 Input 0	0*16&
Matrix Tie Command None Output 1 Input 16	16*1&
Matrix Tie Command None Output 16 Input 16	16*16&
Matrix Tie Command None Output All Input 16	16*1&16*2&16*3&16*4&16*5&16*6&16*7&16*8&16*9&16*10&16*11&16*12&16*13&16*14&16*15&16*16&16*17&16*18&16*19&16*20&16*21&16*22&16*23&16*24&16*25&16*26&16*27&16*28&16*29&16*30&16*31&16*32&16*33&16*34&16*35&16*36&
Matrix Tie Command None Output 36 Input 0	0*36&
Matrix Tie Command None Output 1 Input 36	36*1&
Matrix Tie Command None Output 36 Input 36	36*36&
Matrix Tie Command None Output All Input 36	36*1&36*2&36*3&36*4&36*5&36*6&36*7&36*8&36*9&36*10&36*11&36*12&36*13&36*14&36*15&36*16&36*17&36*18&36*19&36*20&36*21&36*22&36*23&36*24&36*25&36*26&36*27&36*28&36*29&36*30&36*31&36*32&36*33&36*34&36*35&36*36&
Preset Recall 1	1.
Preset Recall 32	32.

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Preset Save 1	1,
Preset Save 32	32,
Refresh Matrix 1 – 16	w0*1*1VC\x0D
Refresh Matrix 17 – 32	w0*17*1VC\x0D
Refresh Matrix 17 – 20	w0*17*1VC\x0D
Refresh Matrix 33 – 36	w0*33*1VC\x0D
Refresh Matrix All	w0*1*1VC\x0Dw0*17*1VC\x0Dw0*33*1VC\x0D
Video Mute Off Output 1	1*0B
Video Mute On Output 1	1*1B
Video Mute Off Output 32	32*0B
Video Mute On Output 32	32*1B
Video Mute Off Output 20	20*0B
Video Mute On Output 20	20*1B
Video Mute Off Output 16	16*0B
Video Mute On Output 16	16*1B
Video Mute Off Output 36	36*0B
Video Mute On Output 36	36*1B

Appendix B. Update Commands

EDID Assignment Input 1	wA1EDID\x0D
EDID Assignment Input 32	wA32EDID\x0D
EDID Assignment Input 20	wA20EDID\x0D
EDID Assignment Input 16	wA16EDID\x0D
EDID Assignment Input 36	wA36EDID\x0D
Executive Mode	X
Input Signal Status Input 1	0LS
Input Signal Status Input 32	0LS
Input Signal Status Input 20	0LS
Input Signal Status Input 16	0LS
Input Signal Status Input 36	0LS
Input Tie Status Input 1 Output 1	w0*1*1VC\x0Dw0*17*1VC\x0Dw0*33*1VC\x0D
Input Tie Status Input 36 Output 36	w0*1*1VC\x0Dw0*17*1VC\x0Dw0*33*1VC\x0D
Output Tie Status Output 1	w0*1*1VC\x0Dw0*17*1VC\x0Dw0*33*1VC\x0D
Output Tie Status Output 36	w0*1*1VC\x0Dw0*17*1VC\x0Dw0*33*1VC\x0D
Video Mute Output 1	1B
Video Mute Output 32	32B
Video Mute Output 20	20B
Video Mute Output 36	36B